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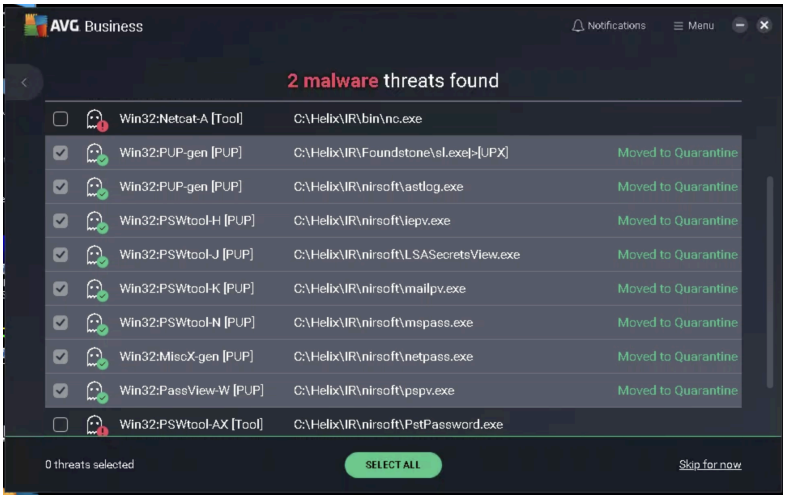
Lab 6: Identifying and Removing Malware on a Windows System

This lab focused on some common procedures to locate and identify threats on a machine using modern antivirus software. We used the AVG tool to first carry out scans to locate threats on our device, then we were able to dig deeper and research the different types of malware and PUPs found, and then finally permanently removed the malicious software and files. We also learned how to quarantine newly discovered threats from ZIP files and how to delete those, as well as set up automated scheduled scans of a system. Additionally, this lab specifically went into a type of ransomware (Cryptcat) and trojan malware (BOClient).

SECTION 1

Part 1: Use Antivirus Software to Scan the Infected System




```
* AVG Scan Report
* This file is generated automatically
*
* Scan name: Deep scan
* Started on: Friday, March 22, 2024 12:00:17 PM
* VPS: 240322-4, 03/22/2024
*

C:\Helix\IR\bin\cryptcat.exe [L] Win32:Malware-gen (0)
File was successfully moved to Quarantine...
C:\Helix\IR\bin\nc.exe [L] Win32:Netcat-A [Tool] (0)
C:\Helix\IR\Foundstone\sl.exe|>[UPX] [L] Win32:PUP-gen [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\astlog.exe [L] Win32:PUP-gen [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\iepv.exe [L] Win32:PSWtool-H [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\LSASecretsView.exe [L] Win32:PSWtool-J [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\mailpv.exe [L] Win32:PSWtool-K [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\mspass.exe [L] Win32:PSWtool-N [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\netpass.exe [L] Win32:MiscX-gen [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\pspv.exe [L] Win32:PassView-W [PUP] (0)
File was successfully moved to Quarantine...
C:\Helix\IR\nirsoft\PstPassword.exe [L] Win32:PSWtool-AX [Tool] (0)
C:\Helix\IR\nirsoft\rdpv.exe [L] Win32:PSWtool-V [PUP] (0)
File was successfully moved to Quarantine...
Infected files: 12
Total files: 256534
Total folders: 22494
Total size: 38.3 GB
```

Screenshots 1 and 2: Contents of the Deep scan file

Document the threat details for the first generic malware threat reported by AVG:

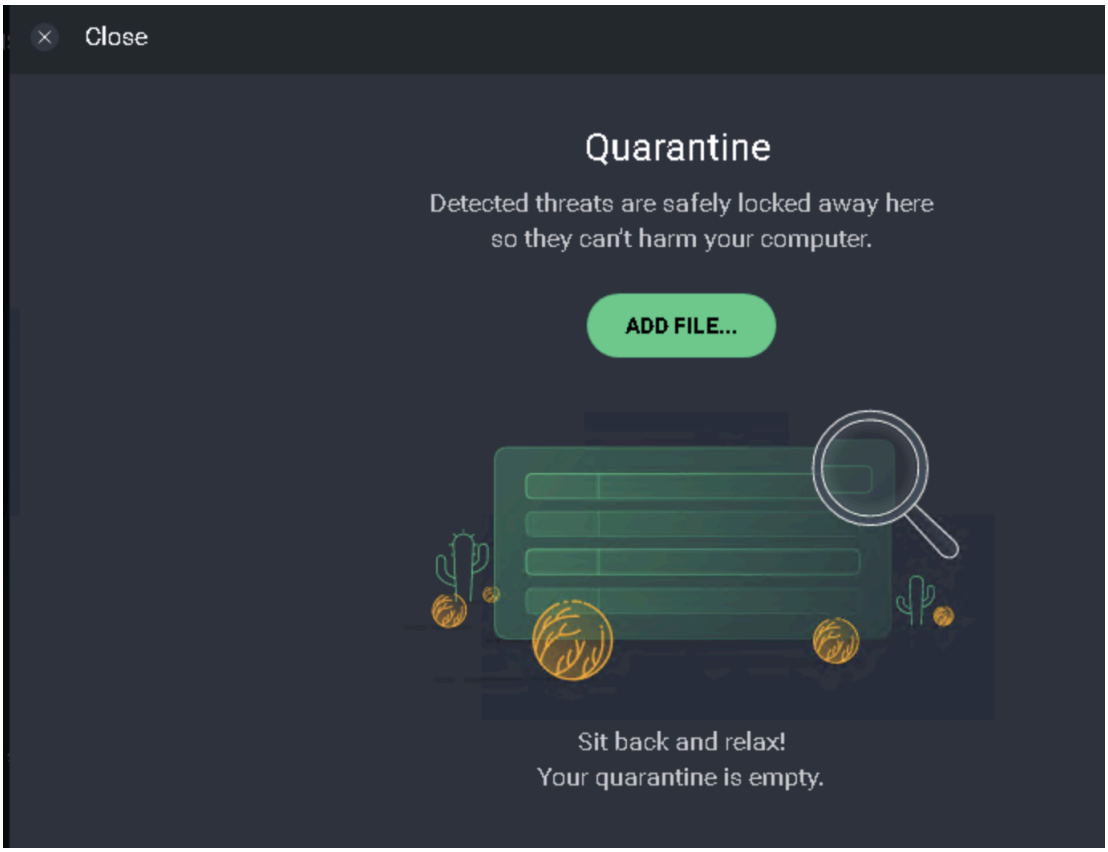
Win32:Malware-Gen from infected file C:\Helix\IR\bin\cryptcat.exe. Cryptcat is a type of ransomware that, when launched, will encrypt files and alter their names. A ransom note will be created on the machine outlining the payment process in which the files will be decrypted. It is believed that decryption without the attacker's assistance is usually impossible. However, in many cases, even after payment, victims still do not receive a decryption key.

Part 2: Identify Threats in Encrypted Archive Files

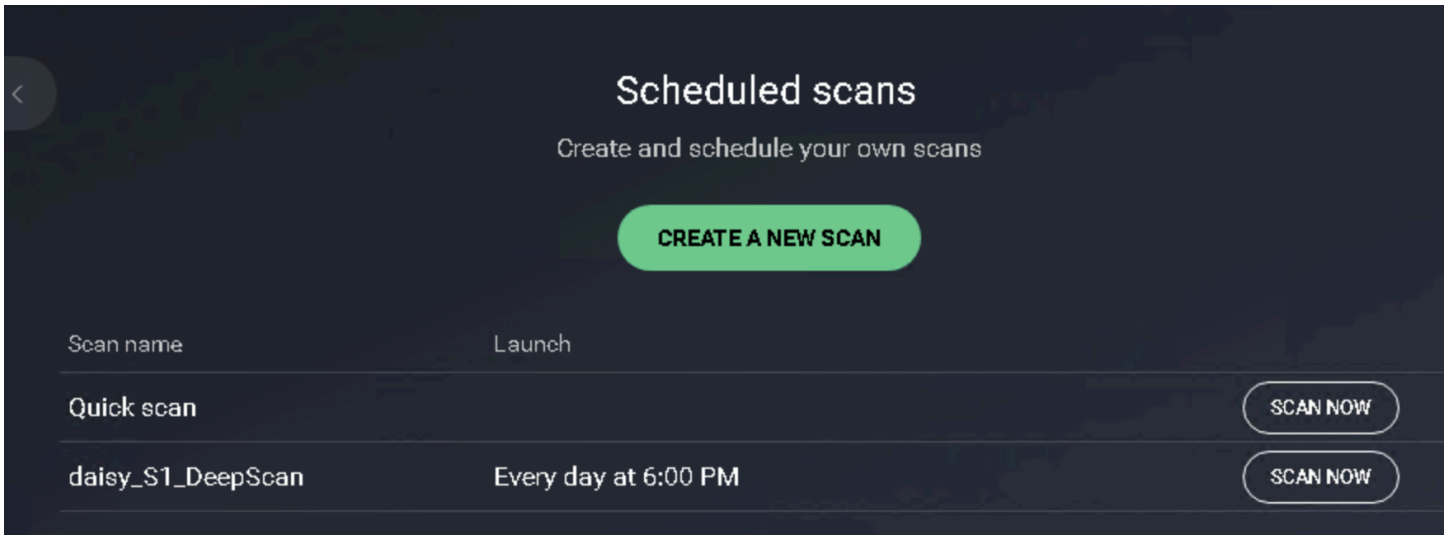
```
* AVG Real-time Shield Scan Report
* This file is generated automatically
*
* Started on: Friday, March 22, 2024 11:56:53 AM
*

3/22/2024 12:39:32 PM   C:\ISSA_TOOLS\prodrev\productreview.pdf [L] JS:Pdfka-FC [Expl] (0)
File was successfully moved to Quarantine...
```

Screenshot 3: Pdfka-FC threat detection in the FileSystemShield file

Part 3: Manage AVG Scans and the Quarantine Area



Screenshot 4: Empty Quarantine area (Virus Vault)



Screenshot 5: Scheduled scan

SECTION 2
Part 1: Use Antivirus Software to Scan the Infected System

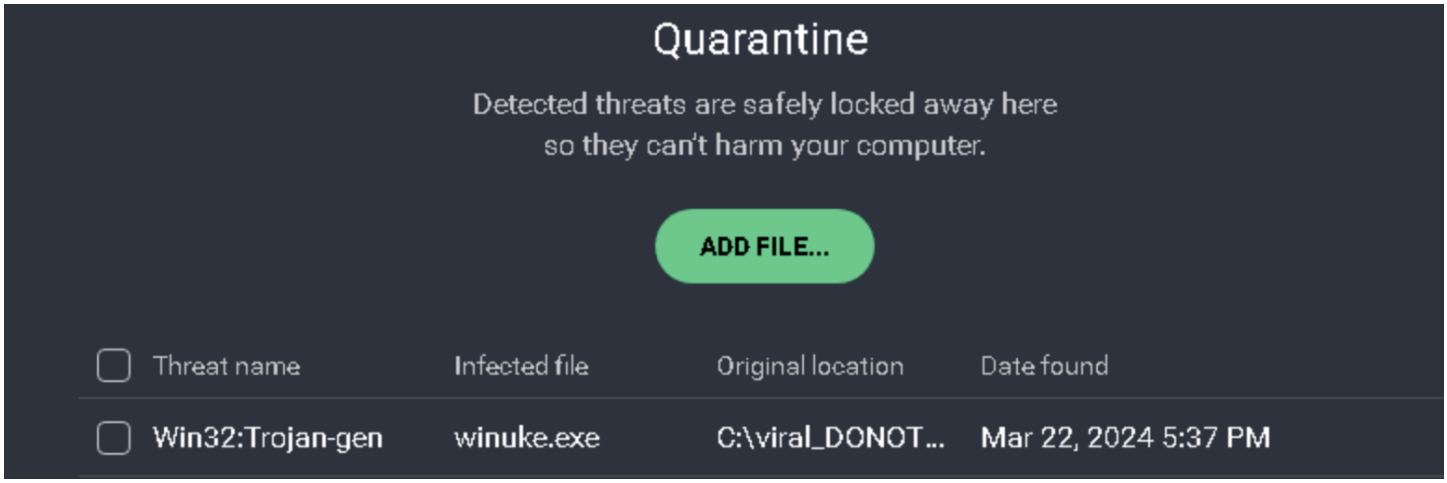
Document the threat details for the first Trojan threat reported by AVG:

Win32:BO-C [trj] located in C:\Users\Administrator\AppData\Local\1\BO\BOCLIENT.exe. This is a program designed for remote system administration. It enables remote control of a computer. The server can be installed without user interaction. It also contains a simple GUI which makes it attractive for inexperienced attackers to use. BO stands for Back Orifice. Attackers can view and modify files on the victim machine, create a log file of user actions, take and send screenshots of the screen, and persist if the computer is rebooted.

```
* AVG Scan Report
* This file is generated automatically
*
* Scan name: Deep scan
* Started on: Friday, March 22, 2024 3:39:36 PM
* VPS: 240322-4, 03/22/2024
```

Screenshot 6: Summary information at the top of the Deep scan file

Part 2: Identify Threats in Encrypted Archive Files

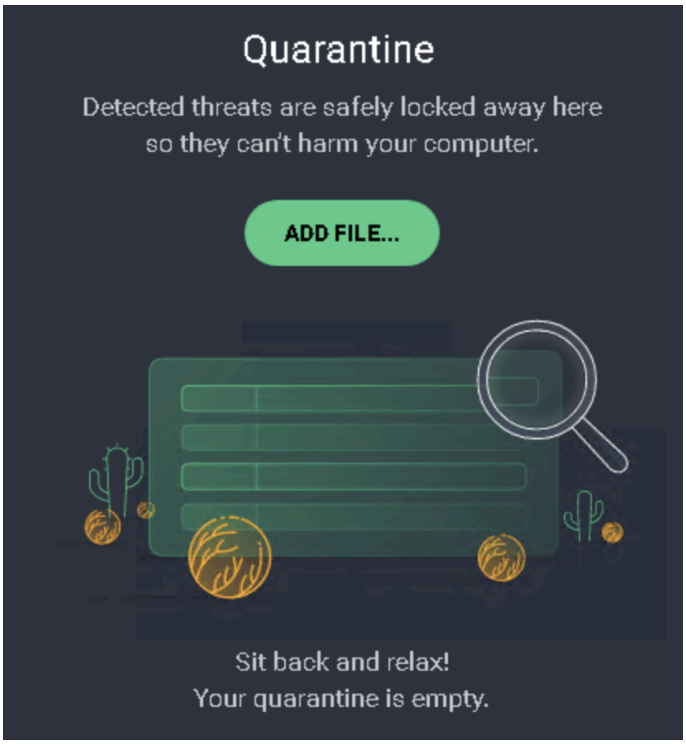


Screenshot 7: Threat details

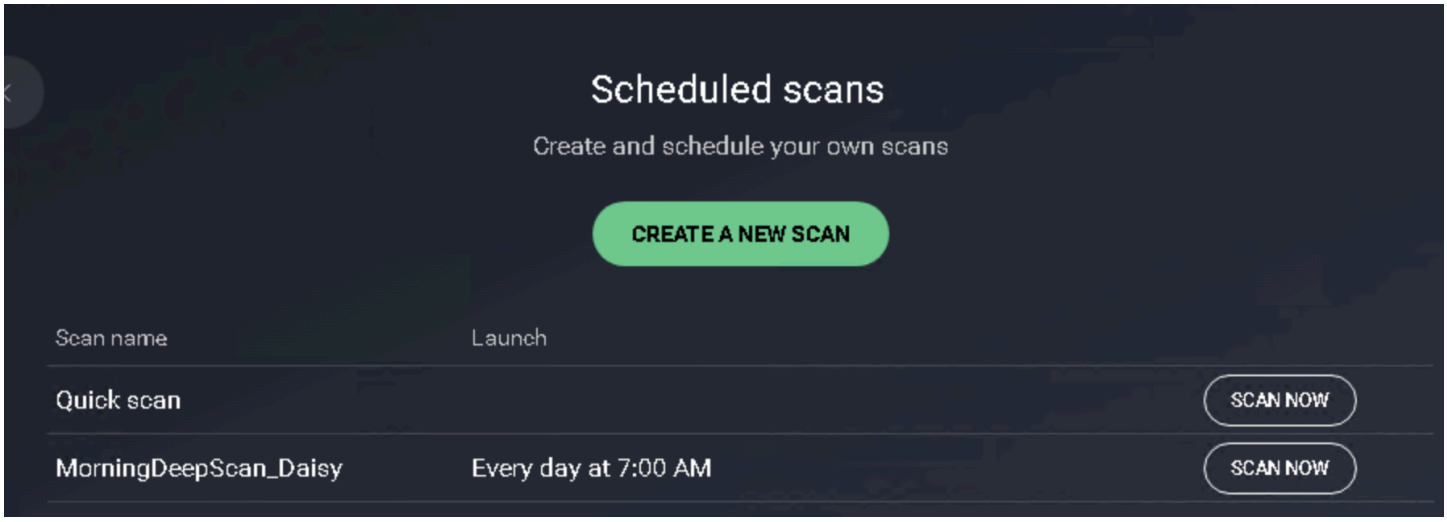
```
* AVG Real-time Shield Scan Report
* This file is generated automatically
*
* Started on: Friday, March 22, 2024 5:00:25 PM
*

3/22/2024 5:37:20 PM    C:\viral_DONOTTOUCH\winuke\winuke\winuke.exe [L] Win32:Trojan-gen
File was successfully moved to Quarantine...
```

Screenshot 8: Winuke threat detection in the FileSystemShield file

Part 3: Manage AVG Scans and the Quarantine Area



Screenshot 9: Empty Quarantine area (Virus Vault)



Screenshot 10: Scheduled Scan